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Assignment No: 7

Problem Statement: Write a C program to accept student details and display their result using an array of structures.

Objectives:
- To understand the concept of structures in C
- To learn how to store multiple records using an array of structures.

Theory: In C programming, a structure is a user-defined data type that allows grouping of variables of different data types under a single name. When multiple records such as student details need to be stored an array of structures is used.

Platform: Linux.

Input: Student roll number, name, and marks in 3 subjects.

Output: Displays students details along with total marks and percentage.

Sample Input: Enter a number of student : 1
Roll no: 101
Name: Rahul.
Marks in 3 subjects: 80 75 90.

Sample Output: Student Result:
Roll No: 101 Name: Rahul Total Marks: 245

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Percentage: 81.67%

Conclusion: Thus, the C program successfully accepts student details using an array of structures and displays the calculated result.

FAQs:

1) What is a structure in C?
A. A structure in C is a user-defined data type that allows you to group different types of data together under one name.

2) What is the difference between a structure and a union in C?
A.

Feature.	Structure.	Union.
Memory	Each member gets seperate memory	All members share same memory.
Size	Sum of all members	Size of largest member
Usage.	When all data is needed	When memory saving is important
Access	All members can be used at same time	Only one member is valid at a time

3) What is an array of structures?
A. An array of structures is a collection of a structure variables stored in an array.
Ex:
struct Student s[3];
This stores details of 3 students.

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4) Why are structures used in C programming?

A. a) They organize related data together
b) Make programs clear and manageable.
c) Useful for handling real-world entities.
d) Allow combining different data types.

5) Advantages of using an array of structures.

A. a) Stores multiple records efficiently.
b) Easy to process data using loops.
c) Reduces code repetition.
d) Better for data management & scalability.
e) Useful in applications like student databases, employee systems.